

PIPELINE REPORT

How DeciWaves works

DeciWaves

DeciWaves reads the Death Stranding, Horizon Zero Dawn Remastered, and Horizon Forbidden West installs you already own and turns their recorded dialogue into MP3 reels you can listen to like an audiobook. Every line carries its speaker and on-screen subtitle where the game data allows, the lines are arranged in story order, and the files are packed to stay under 290 MB each. The tool never modifies the install and ships no game content of its own.

Covers all three game pipelines at the level of their shared shape, with per-game detail where they genuinely differ. Read at commit 3609dd9 on 15 July 2026. The test suite, the CLI help, and the doctor preflight were run on this machine the same day; nothing that costs GPU hours or touches a game install was executed.

Generated 15 July 2026.

Read the code, not just this. Where the two disagree, the code is right.

1.

What this is

A Decima-engine game hides thousands of recorded voice lines inside proprietary archive files, playable only in the order the game decides. DeciWaves is a Windows-only Python command-line tool that reads those archives from a legally owned install, works out which audio clip voices which line of dialogue, and renders the result as story-ordered MP3 reels, each with a plain-text tracklist. The finished product per game is a spreadsheet of voice lines plus a folder of reels you can drop on a phone.

The three supported games share one engine family but ship their dialogue in three different ways, so the codebase deliberately shares only what is genuinely common: the archive readers, the selection and ordering rules, and the MP3 packing at the end. The front half, which is the hard part, is solved three different ways: Death Stranding's data names each line's audio directly; Zero Dawn's does not, so clips are matched to lines by a size fingerprint plus speech recognition; Forbidden West ships a positional index that is replayed to find each clip, then labeled with the game's own subtitle text.

2.

At a glance

1

Setup and preflight

Fetch the three decode tools, record game paths, check readiness



2

Catalog

List every voice line: ID, scene, speaker, subtitle, where its audio hides



3

Bind

Tie each clip to the line it voices; the per-game hard part (GPU for HZD, FW)



4

Order

Arrange lines in story order, with optional user-supplied reference texts



5

Render

Decode to WAV, insert gaps, encode MP3 reels with tracklist sidecars

Each game has a `deciwaves <game> run` command that chains its stages end to end, writing a marker file after each stage succeeds so an interrupted run resumes where it stopped. Every stage can also be run alone with its own flags. A bare `deciwaves` starts a guided interactive mode that asks which game and where to put the output, then runs the identical chain.

3.

Stage by stage

Setup and preflight

`deciwaves setup` pulls three decoder downloads from GitHub into a local tools folder and records the install path for each game you name. `deciwaves doctor` re-checks everything and names the fix for anything missing; a game you simply don't own reports "not configured" and never fails the check. Every later command quietly puts the tools folder on the search path before loading any stage code, because the stage code fixes its tool paths at load time. Both commands ran cleanly on this machine.

IN	command-line flags naming the game install paths
DOES	downloads vgmstream-cli, VgAudioCli, and ffmpeg; finds the Oodle DLL; writes <code>config.json</code>
OUT	%LOCALAPPDATA%\DeciWaves\tools and <code>config.json</code>
NEEDS	internet access to github.com; the game installs on disk
FILES	<code>cli/setup.py</code> <code>cli/doctor.py</code> <code>cli/config.py</code> <code>cli/main.py</code>

Catalog

The output is one spreadsheet row per voice line. Death Stranding's dialogue is described by structured records inside its archives, parsed with a bundled reader; the README puts the result near 27,000 rows. Zero Dawn ships no table of contents at all, so its stage finds dialogue by scanning the archive content and parses it with a small purpose-built byte reader. Forbidden West's `extract` stage walks the game's own streaming index and decodes each English clip to WAV in the same pass, so for that game the audio exists on disk before anything is labeled. All three append to their output as they go and keep a sidecar list of finished work, so a re-run skips what is already done, and a single unreadable file is logged and skipped rather than ending the run.

IN	the game's archive files, read-only
DOES	identifies every voice line and where its audio lives
OUT	<code>out/catalog.csv</code> (DS), <code>out/hzd/catalog.csv</code> , <code>out/fw/clip-index.csv</code> plus decoded WAVs (FW)
NEEDS	DS: the Oodle DLL from the install; FW: VgAudioCli
FILES	<code>engine/catalog.py</code> <code>games/hzd/catalog.py</code> <code>games/hzd/inventory.py</code> <code>games/fw/extract.py</code>

Bind

Death Stranding needs no bind stage: its records name the audio stream per line directly. Story cutscenes are the one exception; their audio arrives as a few whole-scene tracks per cutscene, resolved from a bundled, pre-computed list.

Zero Dawn's clips are anonymous, so every line and every clip is reduced to a cheap fingerprint, the pair of encoded byte length and decoded sample count. When exactly one line and one clip share a fingerprint they bind for free, which covers the large majority. Only the collisions go to GPU speech recognition, where the transcript is fuzzily matched against the candidate subtitles. That pass checkpoints after every clip, heals a half-written checkpoint row before appending to it, and aborts early if the first five clips all fail, which signals a broken environment rather than bad data.

Forbidden West's labels are the game's own subtitle text, read verbatim from each dialogue group. Because a group's subtitle order and audio order are shuffled independently, a local transcription pass decides which subtitle belongs to which clip inside the group; the shipped label is still the exact game text, never the transcript. An optional user-supplied gamescript then adds speaker names and story positions for the lines it covers.

IN	the catalog plus the game's audio clips
DOES	decides which clip voices which line
OUT	a bind manifest CSV per game
NEEDS	HZD and FW: the [asr] install extra and a CUDA GPU; FW: a user-supplied <code>types.json</code>
FILES	<code>games/hzd/binding.py</code> <code>games/hzd/asr_bind.py</code> <code>games/fw/subtitle_bind.py</code> <code>games/fw/subtitle_match.py</code>

Order

Shared selection rules drop lines with no subtitle or no audio and collapse repeats of the same line within one scene. Death Stranding then orders scenes by episode heuristics, sharpened by an optional transcript that anchors scenes to their real chronological position; without one it falls back cleanly. Zero Dawn's quest names carry their own numbering, so its story spine sorts directly, with side content interleaved at its unlock point. Forbidden West orders by gamescript position where the script reaches, groups the unmatched remainder into scene-clustered blocks, and puts the DLC last as an epilogue.

IN	the catalog or bind manifest
DOES	filters, de-duplicates, and arranges lines in story order
OUT	<code>out/playlist.csv</code> (DS); ordering embedded in the manifests for HZD and FW
NEEDS	optional user-supplied reference texts (a DS narrative transcript, the FW gamescript)
FILES	<code>engine/selection.py</code> <code>engine/story_order.py</code> <code>games/hzd/render.py</code> <code>games/fw/story_full.py</code>

Render

Decoded WAVs are cached so re-runs are cheap. Clips get a 0.4-second gap between lines and 1.5 seconds between scenes; long dead air inside cutscene tracks is trimmed. Every input is first re-encoded to one common layout, because the concatenation step would otherwise play mismatched clips at the wrong speed. Whole episodes are packed into files sized to land near 285 MB at the chosen bitrate, then ffmpeg concatenates and encodes each file, and a tracklist CSV with timestamps is written beside it. Per-clip decode failures are logged and skipped; if every attempted clip fails to decode, the DS render exits with an error instead of writing silent, empty reels.

IN	the ordered playlist or manifest, plus the game archives for decode
DOES	decodes clips to WAV, inserts gaps, packs and encodes MP3 reels
OUT	numbered .mp3 reels, each with a <code>.tracklist.csv</code> , under <code>out/ ... /audio</code> or <code>out/fw/reels</code>
NEEDS	ffmpeg; DS: vgmstream-cli and the Oodle DLL; HZD: VGAudioCli
FILES	<code>engine/render.py</code> <code>engine/audio_clip.py</code> <code>games/hzd/render.py</code> <code>games/fw/render.py</code>

4.

The map

<code>src/deciwaves/cli/</code>	The <code>deciwaves</code> command: setup, doctor, guided mode, and the per-game run chains with their resume markers.
<code>src/deciwaves/engine/</code>	The shared pipeline: DS catalog, selection rules, story ordering, clip decode, MP3 packing.
<code>src/deciwaves/engine/pack/</code>	Byte readers for each archive format: DS <code>.bin</code> archives (Oodle-compressed), the HZD/FW package format, and FW's streaming index.
<code>src/deciwaves/games/</code>	Per-game stages in <code>ds/</code> , <code>hzd/</code> , and <code>fw/</code> : what each game needed that the others did not.
<code>src/deciwaves/_vendor/</code>	A bundled, patched MIT-licensed <code>pydecima</code> , the reader for Death Stranding's resource records.
<code>src/deciwaves/data/</code>	Small packaged inputs: the DS file list, pre-resolved cutscene tracks and trim spans, the FW character-name roster.
<code>.memories/</code>	One-fact-per-file notes recording what was reverse-engineered about each format and why.
<code>out/</code> (in your workspace)	Everything generated: catalogs, manifests, WAV caches, error logs, resume markers, and the finished reels. Safe to delete; it is all re-derivable, though slowly.

5.

How to run it

The `--workspace` flag (default: the current directory) decides where `out/` lands; the tool changes into that directory before any stage runs, so every relative path in the pipeline follows it.

1. Install from a clone (not on PyPI)

```
git clone https://github.com/prekabreki/DeciWaves
cd DeciWaves
pip install .
```

For Zero Dawn and Forbidden West also run: `pip install "[asr]"` plus a PyTorch build matching your CUDA version.

2. Fetch tools and register every game you own, in one call

```
deciwaves setup --ds-install "C:\...\DEATH STRANDING DIRECTORS CUT" --
hzd-package "..." --fw-package "..."
```

Name all your games in a single setup run: each run rewrites the whole config, so a later run naming only one game erases the others (finding 2).

3. Check readiness

```
deciwaves doctor
```

Exit code 0 means every required tool is present. Games you don't own show as not configured and never fail the check.

4. Run a game end to end

```
deciwaves --workspace D:\deciwaves ds run
```

Same shape for hzd run and fw run. DS needs no GPU and renders the main-story reel; the full DS reel means running `deciwaves ds render` yourself without `--main-story`. FW needs a `types.json` in the workspace root, and stops cleanly after its subtitle stage until you pass `--gamescript`.

5. Resume or redo

```
deciwaves ds run # skips stages whose out/<game>/done-<stage> marker
exists
```

Delete a stage's marker to force it to re-run. Deleting an early stage's marker does not re-run the later stages; delete their markers too (finding 3).

6.

What could bite you

The foundations are careful. The tool is strictly read-only against the install; resume markers are deliberately separate from outputs, so a crash mid-stage never looks finished; the long GPU pass checkpoints per clip and repairs its own checkpoint file; the render refuses to claim success when nothing decoded. The test suite passed on this machine on

a clean checkout: 456 passed, 33 skipped, in 24 seconds. The findings below are the loose bolts I could verify in the code, ranked by what happens if they go wrong.

The Zero Dawn pipeline quietly transcribes only about 300 ambiguous clips

HIGH

FIX: S

`src/deciwaves/games/hzd/asr_bind.py:94`, `src/deciwaves/cli/run.py:194`

RISK The bind stage's `--sample-cap` defaults to 300, flagged in the code as an MVP cap, and the chained `hzd run` never overrides it. Ambiguous clips beyond the first 300 or so are never transcribed, never bound, and simply missing from the finished reel, while every stage reports success. Nothing in the output says work was capped. The README meanwhile describes this stage as running for hours over the full library, so the docs and the code disagree. Once bind succeeds, its resume marker makes the capped result permanent for that workspace.

FIX Have the `run` chain pass an explicit cap of 0 (or make 0 the default), and print how many buckets were left untranscribed whenever a cap applies.

Registering a second game erases the first game's setup

MEDIUM

FIX: S

`src/deciwaves/cli/setup.py:164`, `src/deciwaves/cli/config.py:21`

RISK `setup` writes all five config keys from this run's flags alone, and unnamed games default to empty. Run `deciwaves setup --ds-install ...` and later `deciwaves setup --hzd-package ...`, and the DS install path and the Oodle DLL path are blanked. The next `ds run` refuses to start, claiming no DS install is configured. The damage is visible but confusing, and the README's own instructions invite exactly this sequence.

FIX Merge the new flags over the existing saved config instead of replacing it, so an omitted flag means "keep what I had".

Re-running an early stage never refreshes the later ones

MEDIUM

FIX: S

`src/deciwaves/cli/run.py:66`

RISK Resume markers only ever skip work; nothing invalidates them. If you delete the catalog marker to rebuild after a game patch, the order and render markers still stand, so the run rebuilds the catalog, skips the rest, and exits successfully while the reels on disk still reflect the old data. A stale result and a fresh one look identical.

FIX When a stage actually executes, delete the markers of every stage after it in the chain; or at minimum say in the skip message that downstream stages were not refreshed.

A mistyped gamescript path ends the Forbidden West run with a success code

MEDIUM

FIX: S

`src/deciwaves/cli/run.py:306`

RISK `fw run` treats a `--gamescript` path that does not exist exactly like no gamescript at all: it prints "no gamescript configured" and exits 0. A typo in the path silently skips the speaker-matching and render stages, and the message blames the wrong cause. Anything scripted on the exit code believes the full pipeline ran.

FIX If `--gamescript` was given but the file is missing, report that path and exit nonzero instead of falling back to the not-configured message.

One Zero Dawn stage swallows read failures without a trace

LOW

FIX: S

`src/deciwaves/games/hzd/wem_metadata.py:72`

- RISK** Every other stage logs the files it fails on; this one skips unreadable dialogue records with a bare catch-and-continue. Lines lost here get no fingerprint and can never bind. The stage's coverage percentage at the end is the only clue, with no way to tell which records were lost or why.
- FIX** Log each skipped record and its reason to an errors file, matching the convention the sibling stages already follow.

The ffmpeg download is not pinned, despite the comment saying it is

LOW

FIX: S

`src/deciwaves/cli/setup.py:33`

- RISK** The setup code's own notes promise release-pinned download URLs so a run next year fetches the same bits. vgmstream and VGAudio are pinned; the ffmpeg URL points at a rolling "latest" build that changes upstream whenever a new nightly lands. A future setup run can fetch an ffmpeg the pipeline was never tested against, and two machines set up a month apart can behave differently.
- FIX** Pin the ffmpeg URL to a dated release tag like the other two tools, and note the pinned version where the others are recorded.

A corrupt config file crashes every command, including the doctor

LOW

FIX: S

`src/deciwaves/cli/config.py:13`

- RISK** Config loading handles a missing file but not a malformed one. A half-written or hand-edited `config.json` makes every `deciwaves` command die with a raw traceback, and `doctor`, the tool meant to diagnose exactly this kind of problem, dies the same way.
- FIX** Catch the JSON parse error, print the config path and a one-line hint to re-run setup, and treat the config as empty.

7.

Open questions

- ? Is the 300-clip transcription cap deliberate for the shipped `hzd run`? The code comment calls it an MVP cap and the README describes an hours-long full pass. I could not determine which reflects current intent; finding 1 assumes the README is the goal.
- ? `games/fw/bulk.py` is imported by nothing except its own test, and `bind.py` with `match_lines.py` look superseded by the subtitle-based path, yet `bind.py` still owns the manifest column list every live FW stage imports. They appear to be leftovers, but nothing confirms they are safe to remove.
- ? Forbidden West has three reel-assembly routes in the CLI (`full-reel`, `weave`, and `dlc` plus `assemble`), and `fw run` uses only `full-reel`. I could not determine whether the other two are supported deliverables or exploratory leftovers.
- ? `ds run` always uses the bundled, pre-resolved cutscene track list rather than regenerating it from the user's install. Whether a future game patch could silently invalidate that bundled data is not something the code checks.

8.

Glossary

Decima The game engine behind all three games; its archive and record formats are what this tool reads.

archive / package The game's own container files, holding thousands of game files inside a few large ones.

resource A structured data record inside a Decima archive; dialogue lines are described by chains of these.

Wwise / .wem A widely used game-audio middleware and its clip container; Death Stranding's voice audio is stored this way.

ATRAC9 Sony's audio compression format; the Horizon games' voice clips use it.

Oodle A compression library that ships as a DLL beside Death Stranding; needed to unpack that game's archives.

transcription (ASR) Turning speech audio into text with a recognition model; here WhisperX on an NVIDIA GPU.

fingerprint The pair of encoded byte length and decoded sample count used to match a Zero Dawn clip to its line without decoding it.

gamescript A fan-compiled transcript of a game's dialogue in play order; user-supplied, never shipped in the repo.

BYO Bring your own: inputs the repo legally cannot ship (game text, the FW type map) that you supply locally.

manifest A CSV listing clips with their labels: line ID, speaker, subtitle, tier, and where the audio is.

resume marker An empty file written after a stage succeeds; its presence is the only thing that lets a re-run skip that stage.

reel One output MP3 file, packed to stay under 290 MB, with a timestamped tracklist beside it.